CivicRecords AI

Open-source, locally-hosted AI that helps American cities respond to open records requests.

CivicRecords AI runs entirely on a single machine inside your city's network — no cloud subscriptions, no vendor lock-in, no resident data leaving the building. It ingests your city's documents, makes them searchable with AI-powered natural language queries, detects potential exemptions, and manages the full request lifecycle from intake to response.

Why This Exists

Every city in America processes open records requests (FOIA, CORA, and state equivalents). Staff manually search file shares, email archives, and databases — then review every document for exemptions before release. It's slow, error-prone, and a growing burden as request volumes increase.

No open-source tool exists for the **responder side** of open records at the municipal level. CivicRecords AI fills that gap.

Key Features

- **AI-Powered Search** — Natural language hybrid search (semantic + keyword) across all ingested city documents with source attribution, normalized relevance scores, and optional AI-generated summaries
- **Document Ingestion** — Automatic parsing of PDF, DOCX, XLSX, CSV, email, HTML, and text files. Scanned documents processed via multimodal AI (Gemma 4) with Tesseract OCR fallback
- **Exemption Detection** — Tier 1 PII detection (SSN, credit card with Luhn validation, phone, email, bank accounts, state-specific driver's licenses) plus per-state statutory keyword matching. Optional LLM secondary review. All flags require human confirmation
- **Request Management** — Full lifecycle tracking with 10 statuses: received, clarification needed, assigned, searching, in review, ready for release, drafted, approved, fulfilled, closed. Timeline, messaging, fee tracking, and response letter generation
- **Guided Onboarding** — Two modes operators can switch between: a 3-phase form wizard (City Profile → Systems → Gap Map), and a single-phase LLM-powered adaptive interview that persists each answer (including `has_dedicated_it`) to the CityProfile singleton and

drives the `onboarding_status` lifecycle (not_started → in_progress → complete). Skip advances the walk truthfully. Both modes surface coverage gaps across 12 municipal domains.

- **Municipal Systems Catalog** — Curated knowledge base of 25+ municipal software vendors across 12 functional domains (finance, public safety, permitting, HR, etc.) with discovery hints and connector templates
- **Universal Connector Framework** — Standardized protocol (authenticate/discover/fetch/health_check) for connecting to city data sources. Ships with four implemented connector types: `file_system` (local/mounted directories), `manual_drop` (watched drop folders), `rest_api` (generic REST API — API key / Bearer / OAuth2 client-credentials / Basic auth; JSON/XML/CSV; page/offset/cursor pagination), and `odbc` (SQL databases via pyodbc, row-as-document with SQL-injection guards). IMAP email, SMB/NFS, and SharePoint connectors on roadmap
- **Scheduled Sync & Idempotent Ingestion** — Per-source cron scheduling (5-field expressions via croniter, evaluated in UTC with local-time disclosure, rolling 7-day min-interval validation, 5-minute floor) with `schedule_enabled` pause toggle. Idempotent pipeline: binary sources dedup by content hash, structured REST/ODBC sources dedup by stable source-path with canonical JSON serialization. Concurrent-update races prevented via `SELECT FOR UPDATE` + partial UNIQUE indexes; content updates atomically replace chunks and embeddings in the same transaction
- **Sync Failure Tracking & Circuit Breaker** — Per-record failure tracking with two-layer retry (task-level exponential backoff + record-level per-tick retry with N=100/T=90s cap). Automatic circuit breaker after 5 consecutive full-run failures (`sync_paused`) with admin-feedback unpause grace period. `health_status` (healthy/degraded/circuit_open) computed live from failure counts. Admin UI: colored health badge, failed records panel with bulk retry/dismiss, Sync Now button with real-time polling progress
- **Operational Analytics** — Real-time metrics: average response time, deadline compliance rate, overdue requests, status breakdown
- **Notification Service** — Template-based notification system with SMTP email delivery via Celery beat (60s interval). Configure SMTP_HOST, SMTP_PORT, SMTP_USERNAME, SMTP_PASSWORD in .env to enable. Notification dispatch into status transitions pending
- **Compliance by Design** — Hash-chained audit logs, human-in-the-loop enforcement, AI content labeling, data sovereignty verification. Designed for Colorado CAIA and 50-state regulatory compliance. CJIS compliance gate for public safety connectors
- **Civic Design System** — Professional UI built with shadcn/ui and civic blue design tokens. Responsive shell: fixed 240px sidebar at ≥768px, hamburger-driven slide-in drawer below that (with focus trap, ESC close, overlay dim, auto-close on route change). WCAG 2.2 AA targeted (44px touch targets, skip-to-content link, icon+color status badges, programmatic label→input associations on admin forms, `role="alert"` validation errors with actionable copy — full third-party accessibility audit still pending)

- **Federation-Ready** — REST API with service accounts enables future cross-jurisdiction record discovery between CivicRecords AI instances

Quick Start

Requirements

- **Docker Desktop** (Windows 10/11, macOS 13+) or **Docker Engine** (Linux)
- **8+ CPU cores**, **32 GB RAM**, **50 GB free disk space**
- No internet connection required after initial setup

Install

> **Install paths.** Two options ship today: > > 1. **Windows double-click installer (T5E, UNSIGNED).** A signed build is a future release — this one is not. The unsigned installer is published on every release tag at `releases/download/<tag>/CivicRecordsAI-<version>-Setup.exe` along with a SHA-256 checksum for independent verification. On first run Windows SmartScreen shows **"Windows protected your PC — Unknown publisher."** This is expected. Click **More info** → **Run anyway** to proceed. See installer/windows/README.md for the full SmartScreen walkthrough and checksum-verify steps. The installer bundles the repo snapshot, runs a prereq check (Docker Desktop, WSL 2 + Virtual Machine Platform, 32 GB RAM floor, optional host Ollama), then runs `install.ps1` (via `installer\windows\launch-install.ps1`). `install.ps1` **auto-pulls `nomic-embed-text` and auto-pulls the Gemma 4 tag you select in the picker** (default `gemma4:e4b`) — expect several minutes on first run — and seeds the T5B baseline datasets. > > 2. **Script-based install (Linux / macOS, and Windows if you prefer CLI).** The scripts below configure and start the Docker Compose stack. They do **not** install Docker, WSL, or any other system prerequisites — those must already be present. `install.ps1` / `install.sh` both ship the 4-model Gemma 4 picker, auto-pull the selected LLM plus `nomic-embed-text`, and auto-seed the baseline datasets on first boot. > > **Two shortcuts, two flows.** The Windows installer creates **separate** Start Menu entries for the two operations — don't confuse them: > > - **Start CivicRecords AI** → daily start. Runs `docker compose up -d` and opens `http://localhost:8080/`. Does **not** run the prereq check, does **not** invoke `install.ps1`, does **not** pull any model, does **not** re-seed data. The Desktop shortcut (if you opted in) mirrors this daily-start behavior. > - **Install or Repair CivicRecords AI** → full bootstrap/repair. Runs the prereq check, then `install.ps1` (which may show the picker and pull models). Use this for first-run setup (the installer fires it automatically for you the first time), when you want to switch LLMs, or to repair a broken stack. > > **Docker Desktop and WSL 2** must be installed and running before either path; the installer detects their absence and prints concrete remediation, but does not install them for you.

Windows: ``powershell git clone https://github.com/scottconverse/civicrecords-ai.git cd civicrecords-ai .\install.ps1`

macOS / Linux: `bash git clone https://github.com/scottconverse/civicrecords-ai.git cd civicrecords-ai bash install.sh`

First Use

1. Open **http://localhost:8080** in your browser 2. Sign in with the admin credentials you configured in .env 3. Go to **Sources** → **Add Source** → enter a directory path to your documents 4. Click **Ingest Now** — documents are parsed, chunked, and indexed automatically 5. Go to **Search** — type a natural language query and get cited results

Phase 1 migration layer

CivicRecords AI backend installs civiccore (the shared CivicSuite schema + migration runtime) as a dependency. During the release-hardening window before PyPI publication, backend/pyproject.toml points at the versioned v0.1.0 GitHub release wheel rather than a Git SHA. This keeps the dependency reproducible without requiring git inside the backend image.

Migrations run in two layers: civiccore first (creates/updates the 16 shared tables), then this repo's Alembic chain on top. See ADR-0003 for the full gate contract.

Architecture

```
` ┌──────────┐ ┌──────────┐ ┌──────────┐ │ Browser │──▶│ nginx
│──▶│ FastAPI │ │ (React UI) │ │ (frontend) │ │ (API) │ └──────────┘
└──────────┘ └──────┬───────┘ │ ┌──────────────────────┤ │ │ │
┌────▼────┐ ┌────▼────┐ ┌────▼────┐ │ PostgreSQL │ │ Redis │ │ Ollama │ │ +
pgvector │ │ (queue) │ │ (LLM) │ └─────────┘ └─────────┘ └─────────┘ │
┌────▼────┐ │ Celery │ │ (worker) │ └─────────┘ `
```

7 Docker services: PostgreSQL 17 + pgvector, Redis 7.2, Ollama, FastAPI, Celery worker, Celery beat, nginx frontend.

Tech stack: Python 3.12, FastAPI, SQLAlchemy 2.0, React 18, shadcn/ui, Tailwind CSS, Alembic, Celery, pgvector, nomic-embed-text, Gemma 4 (recommended).

Configuration

All configuration is via environment variables in .env :

Variable	Description	Default
DATABASE_URL	PostgreSQL connection string	postgresql+asyncpg://civicrecords:civicrecords@postgres:5432/civicrecords
JWT_SECRET	Secret key for JWT tokens	(must be set)
FIRST_ADMIN_EMAIL	Initial admin account email	admin@example.gov
FIRST_ADMIN_PASSWORD	Initial admin account password	(must be set — min 12 chars; installer generates automatically; .env.example placeholder and common defaults rejected at startup)
OLLAMA_BASE_URL	Ollama API endpoint	http://ollama:11434
REDIS_URL	Redis connection string	redis://redis:6379/0
AUDIT_RETENTION_DAYS	Audit log retention period	1095 (3 years)
CONNECTOR_HOST_ALLOWLIST	Comma-separated hostnames/IPs exempt from SSRF block (on-prem use)	(empty — RFC1918, loopback, and cloud IMDS ranges blocked by default)

Supported Platforms

- Windows 10/11 (Docker Desktop)
- macOS 13+ (Docker Desktop)
- Ubuntu 22.04+ / Debian 12+ (Docker Engine)

All platforms use identical Docker containers — the application runs in Linux containers regardless of host OS.

Data Sovereignty

CivicRecords AI is designed for environments where resident data must never leave the network:

- Runs entirely on local hardware — no cloud dependencies
- No telemetry, analytics, or crash reporting
- All LLM inference runs locally via Ollama
- `Verification script confirms no outbound connections:` bash scripts/verify-sovereignty.sh

License

Apache License 2.0 — see LICENSE.

All dependencies use permissive (MIT, Apache 2.0, BSD) or weak-copyleft (LGPL, MPL) licenses. No AGPL, SSPL, or BSL dependencies.

Documentation

Complete System Manual (unified staff + IT admin guide with architecture diagrams):

- Download PDF | Download Word | View Online

Individual References:

- Staff User Manual — For city clerks and records officers (non-technical)
- IT Administrator Manual — Installation, configuration, security, backup, monitoring
- Canonical Spec — Unified Design Specification (single source of truth)
- System Architecture Diagram — Interactive component and data flow diagrams
- Phase Decomposition — Project phases and build sequence

Contributing

See CONTRIBUTING.md for development setup, coding standards, and how to submit changes.

Testing Prerequisites

Integration tests require Docker Compose with PostgreSQL running, and the civicrecords Postgres user needs DROP/CREATE DATABASE privileges (the default docker compose config does this automatically). Each test drops and recreates civicrecords_test to guarantee a clean schema.

`bash docker compose up -d postgres redis docker compose run --rm api python -m pytest tests -q`

If a prior test run died mid-fixture, civicrecords_test may already exist and the suite will fail with DuplicateDatabaseError. Drop it manually and re-run:

`bash docker compose exec postgres dropdb -U civicrecords civicrecords_test`

User Roles

| Role | Permissions | Phase | |------|-------------|-------| | **Admin** | Full access: user management, system config, rule management, audit logs, onboarding | Built | | **Staff** | Search, create requests, attach documents, scan for exemptions, review flags, manage fees | Built | | **Reviewer** | Everything Staff can do + approve/reject responses and exemption flags | Built | | **Read-Only** | View search results and request status only | Built | | **Liaison** | Department-scoped read access to requests and search; Users/Audit Log/Onboarding nav hidden | Built | | **Public** | Submit requests (T5D, authenticated only); track own requests [planned]; search published records [planned] | T5D a57a897 (v1.2.0) |

Service accounts with hashed API keys enable instance-to-instance federation access.

Status

v1.3.0 — 2026-04-25 release. Phase 1 CivicCore extraction landed: civiccore v0.1.0 is now consumed as a release-wheel dependency. Two-layer migration order — civiccore migrations run first via subprocess, then records-side. No API or UI changes (infrastructure only). See CHANGELOG and the v1.3.0 release notes for operator upgrade guidance.

v1.2.0 — 2026-04-23 release. Tier 5 (installer + onboarding + seeding + model picker + portal mode) and Tier 6 (at-rest encryption, ENG-001 closed) ship together. CI green on d556904 (run 24853147133). Backend 617/617 pytest, frontend 36/36 vitest, unsigned Windows installer produced on tag push.

v1.1.0 — Phase 2 release with department access controls, 50-state exemption rules, and compliance templates.

New in v1.1.0:

- Department-level access controls — staff scoped to own department, admins see all
- Department CRUD API with audit logging
- 50-state + DC exemption rule coverage (175 jurisdiction-scoped rules across 51 jurisdictions, plus 5 universal PII rules available in the seed source — 180 total)
- 5 compliance template documents (AI Use Disclosure, Response Letter Disclosure, CAIA Impact Assessment, AI Governance Policy, Data Residency Attestation)
- Template render endpoint with city profile variable substitution
- Exemption auditability dashboard with acceptance/rejection rates and CSV/JSON export (time-period filtering not yet implemented)
- Model registry CRUD endpoints (spec 6.7 compliance metadata)

Carried from v1.0.x:

- 13 staff workbench pages + Login with shadcn/ui design system
- 29 database tables, ~30 API endpoints
- 593 backend + 36 frontend automated tests passing (GitHub Actions CI-verified, run 24817110396 on a57a897)
- Guided onboarding, systems catalog, connector framework
- Request timeline, messaging, fee tracking, response letter generation
- Operational analytics and notification service

- AMD GPU/NPU hardware auto-detection (ROCm on Linux, DirectML on Windows)
- Login rate limiting, audit log archival, admin-only user creation
- Tested on Windows 11 (Docker Desktop) and Ubuntu 22.04 (Docker Engine)

  In v1.2.0 — Tier 5 (all five slices shipped 2026-04-22 → 2026-04-23, tagged in this release):

- **T5C — 4-model Gemma 4 installer picker, 2026-04-22 (7721cf0):** Picker now shows exactly four supported Gemma 4 tags: gemma4:e2b, gemma4:e4b (default), gemma4:26b, gemma4:31b — with per-model disk footprint, min/recommended RAM advisories, and a supportable_against_target boolean against the locked Windows 11 / 32 GB baseline. The fake tags gemma4:12b and gemma4:27b that contaminated install.sh, install.ps1, scripts/detect_hardware.*, and backend/app/config.py have been purged repo-wide. Host RAM is re-verified empirically at install time regardless of picker selection.

- **T5A — Onboarding persistence, 2026-04-22 (1782573):** The single-phase LLM-powered adaptive interview now actually persists each answer onto the CityProfile singleton (creating the row on the first answer), normalizes yes/no → bool for has_dedicated_it, and transitions onboarding_status (not_started → in_progress → complete). Skip advances the walk truthfully (previously it dropped answers silently). Coverage: 4 new persistence tests + 2 skip-truth regression tests; migration 018_city_profile_state_nullable.py relaxes a constraint the persistence path needed.

- **T5B — First-boot baseline seeding, 2026-04-22 (61449c5):** app/main.lifespan now auto-seeds three baseline datasets on first boot: **175 state-scoped exemption rules across 51 jurisdictions** (from STATE_RULES_REGISTRY), **5 compliance templates**, and **12 notification templates**. Idempotent via skip-if-exists — admin customizations survive re-seed. Every run emits a start line, per-dataset created / skipped counts, and a completion summary. Universal PII regex rules are present in the seed source (5 additional rules, 180 total) but are **not** seeded by first-boot because they lack a two-letter state_code; those remain deferred pending schema relaxation. See backend/app/seed/first_boot.py and §8.7 of the spec.

- **T5D — Install-time portal switch (private vs. public), 2026-04-23 (a57a897):** New PORTAL_MODE environment variable (private | public, default private) locked at install time, changeable post-install by editing .env and restarting the stack. The installer (install.ps1 / install.sh) prompts for the choice interactively; non-interactive installs can pre-set

CIVICRECORDS_PORTAL_MODE `. Case and whitespace are normalized by a` field_validator `on the config model. **Private mode (default)** is staff-only — the login screen is the only externally reachable page,` /auth/register `returns 404, and` UserRole.PUBLIC `is not assignable. **Public mode** exposes exactly three surfaces and nothing more: (1) a public landing page, (2) a resident-registration path, and (3) an authenticated records-request submission form for` UserRole.PUBLIC `users. Submission requires authentication — anonymous walk-up submission is intentionally out of scope. Staff roles (ADMIN, STAFF, REVIEWER, READ_ONLY, LIAISON) continue to use` /requests/ `and receive 403 on the public submit endpoint. An unauthenticated` GET /config/portal-mode `endpoint (typed` PortalModeResponse { mode: Literal["public","private"] } `) is always mounted so the frontend can discover the active mode on boot and branch its routing. Fixed a pre-existing bug in` UserCreate `that forced self-registered users to` UserRole.STAFF `; self-registration now correctly forces` UserRole.PUBLIC `(and is still only reachable in public mode). Coverage: 15 pytest cases in` backend/tests/test_portal_mode.py `plus 12 vitest cases across` PublicLanding.test.tsx `,` PublicRegister.test.tsx `, and` PublicSubmit.test.tsx `. Explicitly **not** shipped in this slice and not implied by any copy: published-records search, a full resident dashboard, a track-my-request suite, or any other public-portal feature.`

- **`T5E — Windows unsigned double-click installer, 2026-04-22 (`** **1d5429d** **`; test-harness flake fix`** **e898319** **`):`** `Real Windows` .exe `installer built with Inno Setup 6.x, produced on every` v* `tag by` .github/workflows/release.yml `on` windows-latest `via` choco install innosetup -y `+` installer/windows/build-installer.sh `. **Unsigned by design per Scott-locked B3=α posture** — operators must expect SmartScreen "Windows protected your PC — Unknown publisher" on first run; install path documented in` installer/windows/README.md`. Flow is split into two Start Menu shortcuts: **Start CivicRecords AI** (daily` docker compose up -d `) and **Install or Repair CivicRecords AI** (full bootstrap + picker + model pull). Version is tag-derived (no hardcoded version drift) via` /DMyAppVersion= `from` $CIVICRECORDS_VERSION `. Desktop shortcut mirrors daily-start. macOS and Linux remain on the script path (` install.sh `) — native installer parity on those platforms is explicit follow-on work, not scheduled.`

- **`T3D regen (`** bf3c9c3 `) — Regenerated` docs/openapi.json `and` frontend/src/generated/api.ts `after the T5A schema change; CI's stale-check gate enforces this on every subsequent backend schema or route change.`

- **`CI hygiene (`** 5dbeed7 `) — Bumped` actions/checkout@v4 `and` actions/setup-node@v4 `for Node 24 runtime support ahead of the 2026-06-02 GitHub runner default flip.`

In v1.2.0 — security hardening carried from earlier sprints (all tagged in this release):

- **T2A** — Role self-escalation via PATCH /users/me closed (UserSelfUpdate schema); all 24 department-scoped handlers enforce require_department_scope ; 404/403 status-code info-leak unified via require_department_or_404 ; Pattern D list-endpoint fail-open fixed on 4 routes; parameterized enforcement test covers 25 routes

- **T2B** — connection_config redacted from GET /datasources/ for all non-admin users (DataSourceAdminRead returned only on admin write endpoints). **ENG-001 runtime exposure: closed.**

- **Tier 6 / ENG-001 — At-rest encryption for data_sources.connection_config , 2026-04-23:** Connector credentials are now encrypted at rest using Fernet (AES-128-CBC + HMAC-SHA256) via a transparent SQLAlchemy EncryptedJSONB TypeDecorator; envelope shape is {"v": 1, "ct": "<fernet-token>"} . A new required ENCRYPTION_KEY env var (installer auto-generates on fresh installs) drives the encryption; a reversible Alembic migration (019_encrypt_connection_config) encrypts existing plaintext rows and decrypts on downgrade. Operator verification: docker compose run --rm --no-deps api python scripts/verify_at_rest.py . **ENG-001 now fully closed.** Key rotation is not supported in this release and is tracked as a future slice; the versioned envelope ("v": 1) leaves the door open.

- **T2C** — FIRST_ADMIN_PASSWORD startup validator (rejects .env.example placeholder, empty, <12 chars, and common defaults); SSRF host validator blocks loopback, IMDS, and RFC1918 ranges on REST and ODBC connector URLs at schema-validation time

- **T3A** — Create-user form now POSTs to /api/admin/users (was /api/auth/register , which silently downgraded submitted roles to STAFF)

- **CI** — 593 backend + 36 frontend tests (CI-verified on a57a897 , run 24817110396); bootstrap-failure smoke test confirms stack rejects placeholder admin password; docs/openapi.json and frontend/src/generated/api.ts stale-check enforced on every build (T3D)

`> ENG-001 standing caveat (historical — closed 2026-04-23): At-rest encryption for` data_sources.connection_config `landed as Tier 6 / ENG-001. Prior to Tier 6 the JSONB column was plaintext and visible to any database superuser,` pg_dump `output, or restored backup. Post-Tier-6 the column is stored as a Fernet envelope keyed off` ENCRYPTION_KEY `;` pg_dump` output and raw backups contain ciphertext only. T2B runtime exposure closed in an earlier sprint; Tier 6 closes the at-rest gap. ENG-001 is now fully closed. See Section 8.10 of docs/UNIFIED-SPEC.md and the Operator section on the encryption key in USER-MANUAL.md for details.

Roadmap (per canonical spec):

| Phase | Focus | Key Deliverables | Status | |-------|-------|----------------|--------| | **Phase 0** | Design foundation | shadcn/ui, design tokens, sidebar shell, component library | In progress | | **Phase 1** | Staff workbench redesign | Redesign all staff pages with new design system, WCAG 2.2 AA | In progress | | **Phase 2** | New backend features | Fees, notifications (SMTP), response letters, context manager, analytics, liaison role, department scoping, audit retention | Partially built | | **Phase 3** | Public portal | Public homepage, search, guided request wizard, request tracker, help pages | Partial — T5D minimal surface shipped (landing + resident-registration + authenticated submission); published-records search, resident dashboard, and track-my-request remain Planned | | **Phase 4** | Transparency layer | Open records library, reporting dashboards, public archive, federation | Planned (v2.0) |

Note: Version numbers (semver) track release history. Phase numbers track design completeness per the canonical spec. They are separate systems. Current build (v1.3.0) includes backend work from Phases 0-2 and partial Phase 3 (T5D minimal public portal surface), but has not completed the full scope of any phase. See canonical spec for complete requirements and reconciliation for current gap analysis.